

22 popquiz AC generator ↕

⚠ This is a preview of the published version of the quiz

Started: Mar 23 at 6:42pm

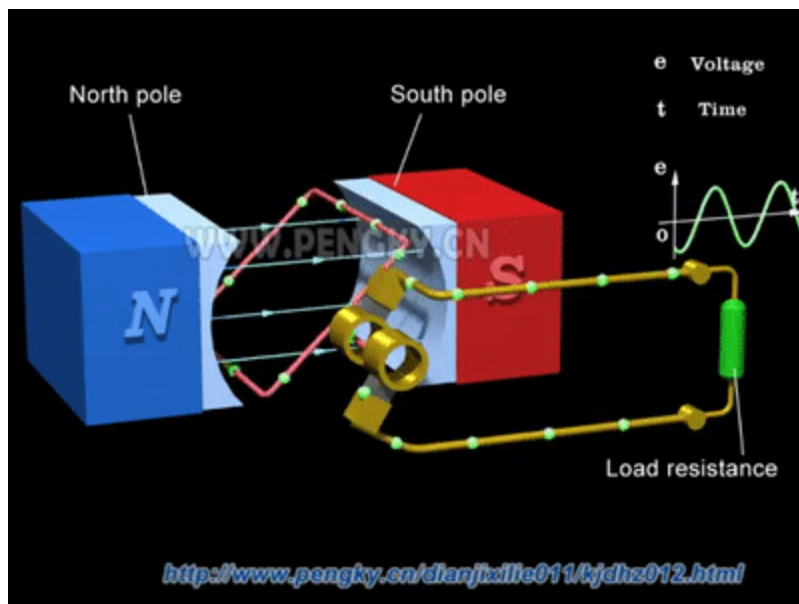
Quiz Instructions

Review:

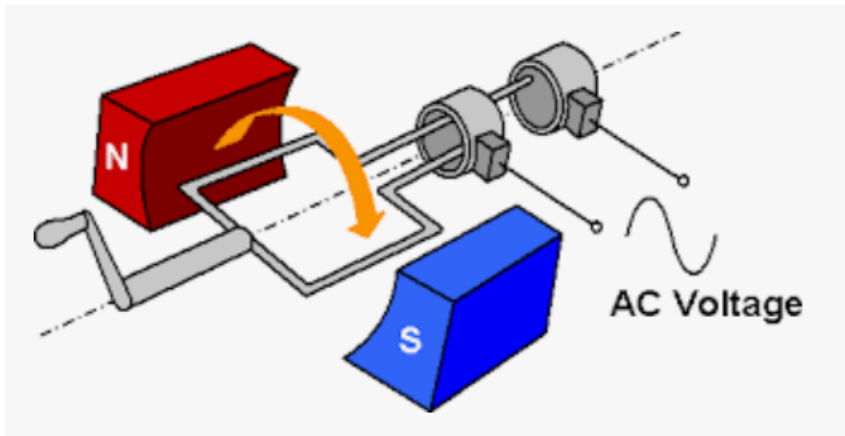
Consider the voltage $V(t)$ changing with time (t). Induced by a AC generator. The voltage follows a sine function. The equation for a sine function is: $V(t) = E_{\max} \sin(\omega t)$ with $\omega = 2\pi f = 2\pi/T$. f is the frequency of the sine function (cycles per second). T is the period of the sine function (seconds per cycle). $T = 1/f$ or $f = 1/T$.

$E_{\max} = NBA\omega$

ω is also how fast the loop is spinning in the magnetic field in rad/s. with $2\pi \text{ rad} = 1 \text{ rotation} = 1 \text{ revolution} = 1 \text{ cycle}$. ω is the rotational speed or the rate of spin of the loop.



Question 1 10 pts



If the loop makes 50 turns per second (50Hz) then the rotational speed ω (rad/s) is about

hint: 1 cycle (one revolution , one turn) = 2π

☐

50 rad/s

☐

100 rad/s

☐

314 rad/s

☐

628 rad/s

☐

Question 2 10 pts

Building up on previous question. The period of the motion is (time for 1 cycle or one turn or one revolution)

☐

2s

☐

0.02s

☐

3.14s

☐

50s

☐

Question 3 10 pts

The magnetic field from the permanent magnet is 0.1 T. The width of the loop is 10cm and its length is 20cm. There are 10 loops.

What is the maximum emf (voltage) induced (see slide 17 $E_{\max}$)



about 1V



about 50V



about 6V



about 3V

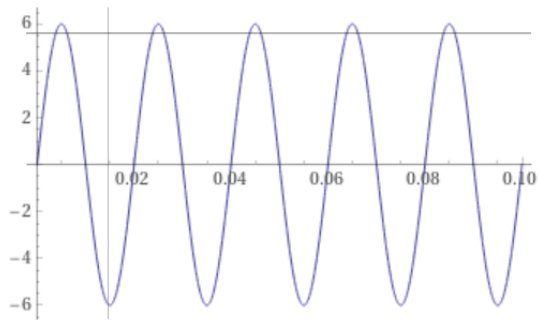


Question 4 10 pts

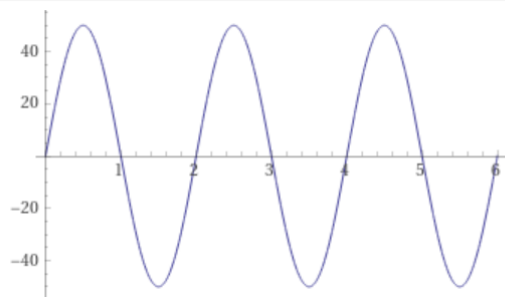
Building up on previous question. What is the graph of the AC voltage generated by the generator?

(slide 17)

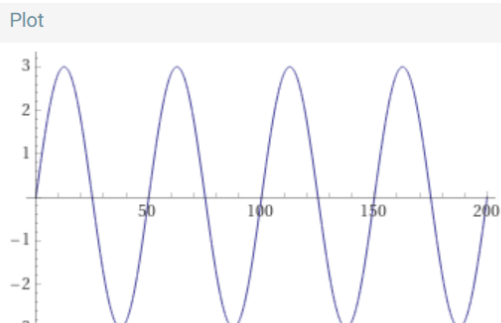
A.



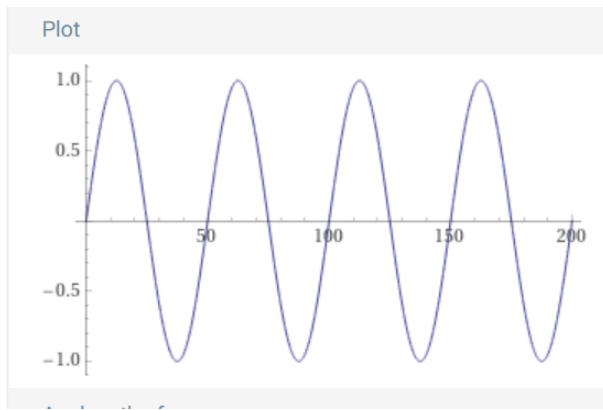
B.



C.



D.

☐

C

☐

A

☐

B

☐

D

⋮

Question 5 10 pts

If the loop goes faster, the voltage

☐

can't be determined

☐

0

☐

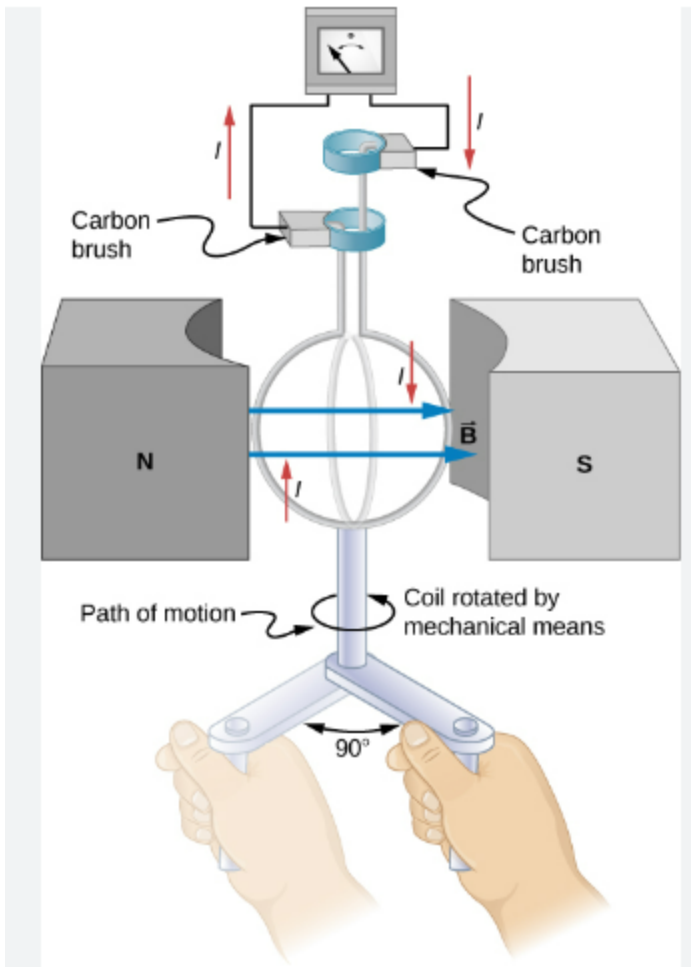
decreases

☐

increases

⋮

Question 6 10 pts



A rotor spins in a permanent magnetic field. Its rotational speed is 30 rot/min. What is the speed in rad/s?

- ☐ 3 rad/s
- ☐ 0.5 rad/s
- ☐ 6 rad/s
- ☐ 30 rad/s



Question 7 10 pts

Building on the previous question. What is the period of the rotor?

- ☐ 0.1s
- ☐ 0.5s



1s



2s



Question 8 10 pts

Building on previous question. What is the frequency



1Hz



2Hz



5Hz



0.5Hz



Question 9 10 pts

Building on the previous question. What is the emfmax induced (so the peak)? (see slide for equation). The magnetic field is 0.1T and the rotor is a circular loop with a radius 10cm (find area A!) . Number of loop is 10.



about 10mV



about 50mV



about 5mV



about 90mV



Question 10 10 pts

Building on the previous question. What is the maximum magnetic flux Φ going through the loop. See definition of flux. (slide 26 big ideas)



0.003 Wb



0.01 Wb



1 Wb



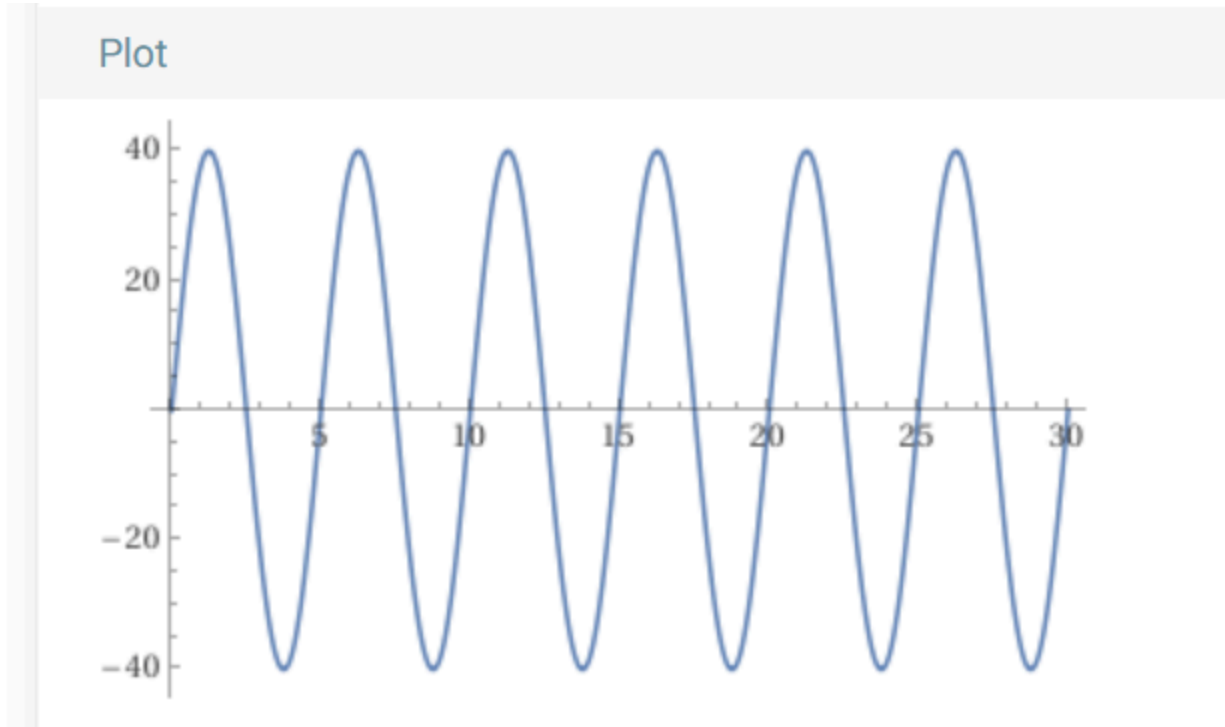
0.03 Wb



Question 11 10 pts

Consider the voltage $E(t)$ changing with time (t). Induced by a AC generator. The voltage follows a sine function. The equation for a sine function is: $E(t) = E_{\max} \sin(\omega t)$ with $\omega = 2\pi f = 2\pi/T$. f is the frequency of the sine function (cycles per second). T is the period of the sine function (seconds per cycle). $T = 1/f$ or $f = 1/T$.

ω is also how fast the loop is spinning in the magnetic field in rad/s. with $2\pi \text{ rad} = 1 \text{ rotation}$.



What is the peak value?

☐ 10

☐ 40

☐ 10

☐ 20

☒

Question 12 10 pts

Building on the same question. What is the period (T) ? that is the time for one cycle.

☐ 5s

☐ 2.5s

☐ 10s



50s



Question 13 10 pts

Building on the previous question. What is the equation of the sine function?

slide 17 for equation

 $5 \cdot \sin(5x)$  $40 \cdot \sin(2\pi x)$  $40 \cdot \sin(2\pi/5x)$  $10 \cdot \sin(2\pi/5)$

Not saved

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